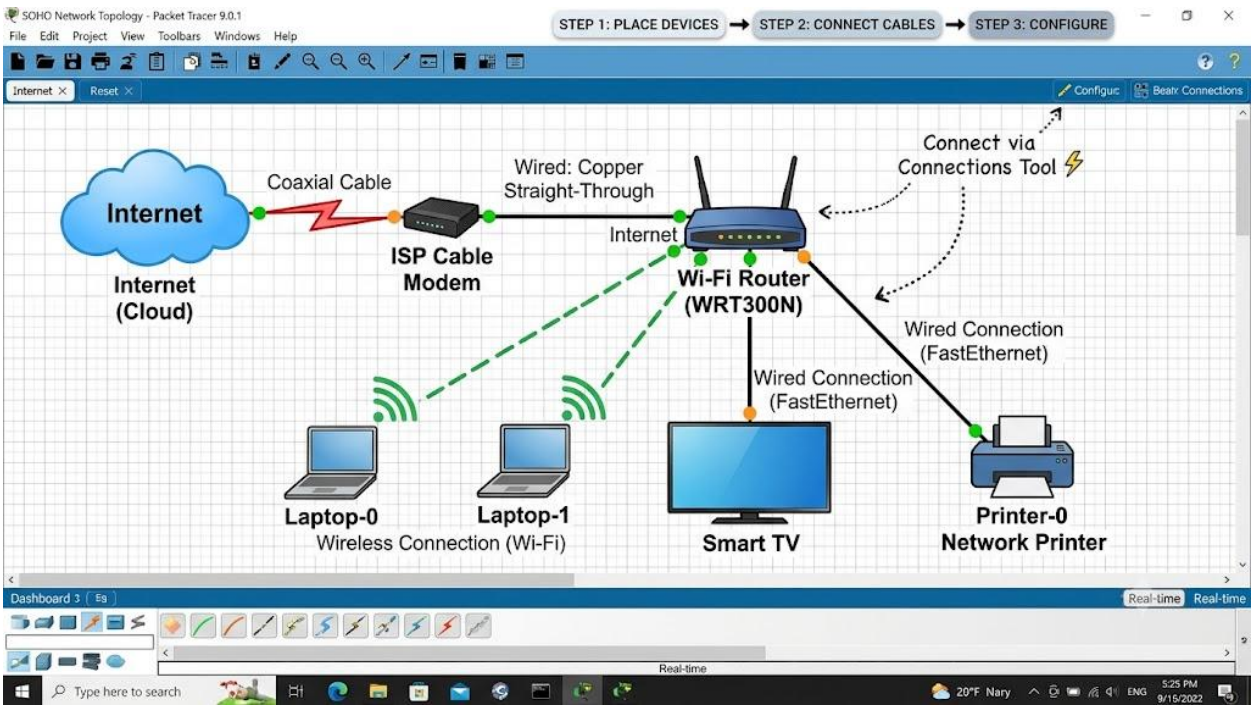
Assignment 1

SOHO Network

1. **Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).**



2. **List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

❖ **Hardware required for the SOHO network**

Component	**Suggested brand/model**
Broadband internet connection	JioFiber or Airtel Xstream Fiber
Wi-Fi router	TP-Link Archer C6
2 laptops	ASUS / HP
Smart TV	Samsung / LG Smart TV
Printer	HP DeskJet 2331
Ethernet cable (RJ45)	Cat6 cable
Power adapters	Included with devices

Note: No extra switch or Wi-Fi extender is needed for this small setup if the router has sufficient ports and Wi-Fi coverage. The laptops and TV can connect wirelessly, and the printer can connect through USB or Wi-Fi if supported.

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

1. Connect the **Wi-Fi router to the broadband connection** and turn it on.
2. Connect my **laptop to the router's Wi-F**i.
3. Open a browser and enter the router's IP address, such as 192.168.0.1.
4. **Log in using** the default admin username and password.
5. Open **Wireless Settings** and set a unique Wi-Fi name (SSID), such as **My_SOHO_Network.**
6. Select **WPA2-PSK (AES) security** and create a **strong Wi-Fi password**.
7. Open System Tools / Administration and **change the default admin password** to a strong, unique password.
8. Click Save/Apply and restart the router if required.
9. Reconnect both laptops and the smart TV using the new Wi-Fi password.
10. Test the internet connection on all connected devices.

4. **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

- **I assigned static IP addresses to the printer and smart TV. The two laptops receive their IP addresses automatically through DHCP from the Wi-Fi router.**

Device	MAC Address (Example)	IP Address	Configuration
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	Static
Laptop 1	00:1A:2B:3C:4D:02	192.168.1.100	DHCP
Laptop 2	00:1A:2B:3C:4D:03	192.168.1.101	DHCP
Smart TV	00:1A:2B:3C:4D:04	192.168.1.10	Static
Printer	00:1A:2B:3C:4D:05	192.168.1.20	Static

5. **Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue**

❖ **If one laptop cannot connect to the internet, I would check the following issues:**

1. Wi-Fi Disconnected: I would check whether Wi-Fi is enabled and connect the laptop to the correct network**.**

2. Incorrect IP Address: I would run ipconfig in Command Prompt. If the IP address is incorrect, I would run ipconfig /renew to request a new address through DHCP.

3. Router Problem: I would check whether other devices can access the internet. If none can connect, I would restart the router and modem.

4. DNS Problem: I would run ping 8.8.8.8 and ping google.com to check internet and DNS connectivity. If DNS fails, I would check the DNS settings.

Result: These steps help identify and fix common internet connectivity problems.